TE WHARE WĀNANGA O TE ŪPOKO O TE IKA A MĀUI
VICTORIA
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SCHOOL OF ECONOMICS
AND FINANCE

ECON412 International Economics: Trade

Trimester 2, 2018

Lecture Week #3: The Heckscher-Ohlin Model

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Overview

- **Readings:** Feenstra Chapter 2 (pages 25-33). KOM Chapter 5.
- Goal is to explain the trade patterns between two countries.
- Two formulations
  - Heckscher-Ohlin-Samuelson (HOS): two by two by two.
  - Heckscher-Ohlin-Vanek (HOV): many by many by many.
- We focus on the theory. Empirical tests are left for self-reading.
- Main problem: model fails on empirical tests.

1. Heckscher-Ohlin-Samuelson Model (HOS)

1-1. Key assumptions

- Two goods; two factors; two countries (Home and Foreign).
- Countries are identical in technology, and in preference (homothetic).
- Countries differ in factor endowments.
- Free trade in goods, but not in factors.
- Unit cost functions have one common point.
- These assumptions imply: **factor prices are equalized across countries.**

For the purpose of presentation, in addition, assume that:

- Home is more abundant in labor, $\frac{L}{K} > \frac{L^*}{K^*}$.
- The production of good 1 uses labor more intensively, i.e., $\frac{l_1(w,r)}{k_1(w,r)} > \frac{l_2(w,r)}{k_2(w,r)}$.
- Trade is balanced, e.g., value of exports equal to value of imports.

1-2. Heckscher-Ohlin Theorem

> *Each country will export the good that uses its abundant factor intensively.*

In our case, Home exports good 1, and Foreign exports good 2.

Intuition: By assumption, $\frac{L}{K} > \frac{L^*}{K^*}$. Since good 1 is more labor intensive, if countries are in autarky (no trade), country 1 will produce more amount of good 1 than country 2, implied by the Rybczynski Theorem. The assumption that countries have identical preferences implies that the demand for each good is identical across countries. This in turn implies that relative price in autarky ($p^a = p_1/p_2$) is different between countries, raising the need for international trade. In our case, $p^a < p^{a*}$, Home will export good 1.

Proof of Heckscher-Ohlin Theorem. Suppose that $L = L^*$ and $K < K^*$.

1.--Start with the autarky. The world in equilibrium is in autarky **if** relative price equalizes across countries. Thus, we want to check whether relative price equalizes if we assume that autarky is the equilibrium. In autarky, the equilibrium outputs lie at the tangent point of the production possibility frontier (PPF) and the consumer indifference curve. The consumer theory suggests that at the optimal consumption point, the relative price $p^a = p_1^a/p_2^a$ is the (negative) slope of the indifference curve. Hence the slope at the tangent point between the PPF and the indifference curve is the (negative) value of p^a. See Figure 1.

2.--Two countries differ in the PPF. If relative price is the same between Home and Foreign, then, by Rybczynski Theorem, Foreign produces more of good 2 and less of good 1, both relative to Home.

Two countries have identical preference (which is homothetic), meaning that the relative demand (demand for good 1 relative demand for good 2) is the same in two countries. Had the relative price been the same between two countries, the demand in Foreign is on the same straight line as the demand at Home (Point C' in Figure 1), which is different from what is offered by Foreign producers (Point B').

3.--Therefore, the relative price must differ between two countries if they are in autarky. Had the relative price in autarky been the same, there would have been excess demand for good 1 in Foreign, implying that the autarky prices must satisfy that $p^a < p^{a*}$.

Trade

4.--Since $p^a < p^{a*}$, Home will export good 1 and import good 2, up to a point where relative price equalizes ($p = p^*$). Thus the world relative price p^t satisfies: $p^a < p^t < p^{a*}$. Home produces at point B and Foreign produces at point B^*, in Figure 2. Home consumes at C and Foreign consumes at C^*.

Figure 1.

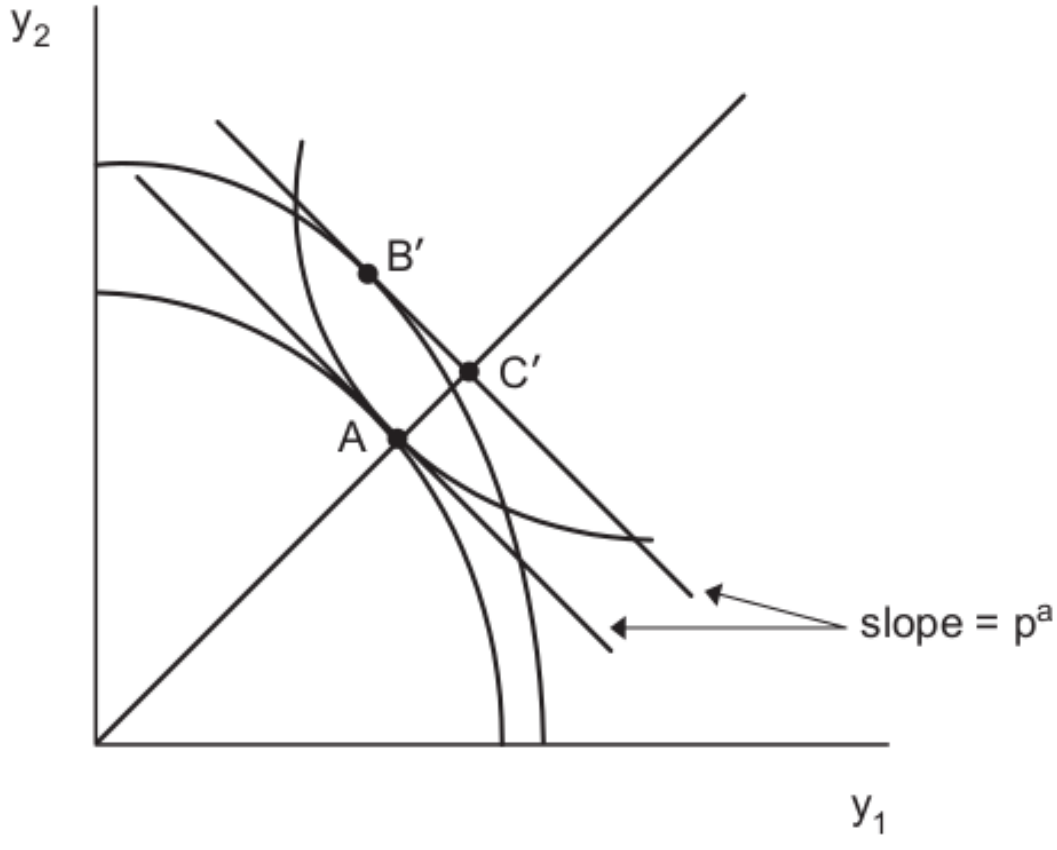
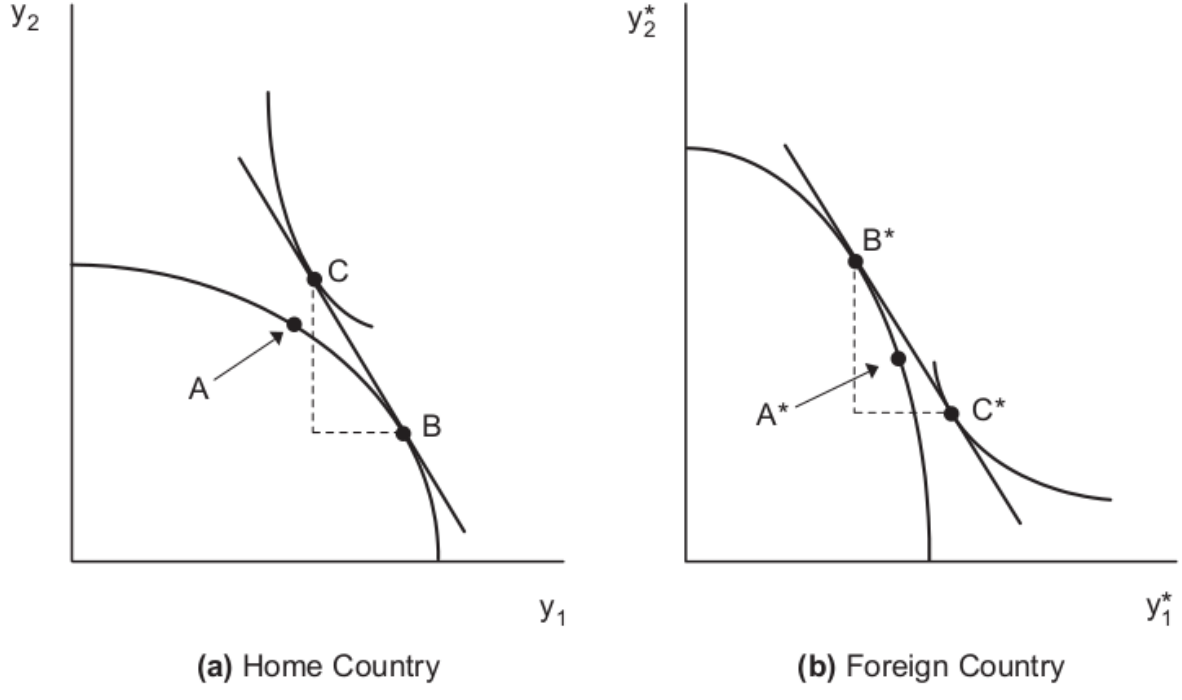


Figure 2.



Distributional effects of trade

- From autarky to free trade, the *abundant* factor gains from trade (in our case, wage at Home rises); The scarce factor losses (in our case, capital return in Foreign declines).
- This is the result of Stolper-Samuelson Theorem. In autarky, the marginal product of the abundant factor (labor) at Home is lower than that in Foreign, so real wage at Home was lower than in Foreign in autarky.
- At Home, following the free trade, real wage rises (because $p^t > p^a$) and real return of capital declines.
- Factor prices are equalized across countries.

2. Heckscher-Ohlin-Vanek Model (HOV)

The goal is to, for a country, relate the factor content of trade to the underlying endowments.

- Many goods; many factors; many countries.
- Same assumptions as in HOS model
  - Technologies are identical across countries, as well as preferences.
  - Production functions are homogeneous of degree 1.
  - Preference is homothetic.
- Free trade implies that factor price is equalized across countries, for all factors.

2-1. Model environment

- Goods indexed by $j = 1, \cdots, N$;
- Factors indexed by k or $l = 1, \cdots, M$.
- Countries indexed by $i = 1, \cdots, C$.

- Let the $M \times N$ matrix $A = [a_{jk}]'$ denote the amounts of factors (k) need to produce one unit of output in sectors (j). A has the shape of $M \times N$.
  - E.g., $A = \begin{bmatrix} l_1(w,r) & l_2(w,r) \\ k_1(w,r) & k_2(w,r) \end{bmatrix}$.

- Let Y^i be a column vector of length N, representing the outputs produced in country i.

- Let D^i be a column vector of length N, representing the demand for goods (used for consumption) in country i.

- The exports in country i are then $T^i = Y^i - D^i$.
- The **factor content of trade** is defined as $F^i = AT^i$, F^i is a $M \times 1$ vector.
- Let p be the column vector of output prices.

2-2. Heckscher-Ohlin-Vanek Theorem (HOV)

- We want to compute F^i as an expression of parameters of technologies and preferences, as well as endowments.
- Note first that $F^i = AY^i - AD^i$, i.e., factor content of trade is demand for factor in country i's production net of its 'demand' for factor embodied in consumption.

- The demand for factors in country i is AY^i (implied by the production functions of homogeneous of degree 1). Let $AY^i = V^i$. E.g., recall the case of two goods and two factors: $l_1(w,r)y_1 + l_2(w,r)y_2 = L$.
- V^i is the endowment vector for country i.

- The demand for goods in country i satisfies $D^i = s^i D^w$
  - D^w is the world consumption vector; s^i is the share of country i in world consumption.
  - Why? Preference is homothetic, and identical across countries.
  - Product prices are equalized across countries, under free trade.
  - This implies, consumption is proportional to each other for all countries: $s^i = \frac{p'D^i}{p'D^w}$. s^i is a scalar.

- It follows then that $AD^i = s^i AD^w$.

- Trade is balanced, **total** expenditure equals to **total** value of production for each country, meaning $p' D^i = p' Y^i$.

- This implies that $AD^i = s^i AD^w = s^i AY^w = s^i V^w$.
  - The second equality holds because, the world total expenditure equals world total production

- **HOV Theorem**: $F^i \equiv AT^i = V^i - s^i V^w$.
  - For factor k, we have $F_k^i = V_k^i - s^i V_k^w$.
- If country i's endowment of factor k relative to the world endowment exceeds country i's share of world GDP, e.g., $V_k^i / V_k^w > s^i$, then country i is *abundant* in factor k.
- HOV Theorem says that the factor content of trade in factor j should be positive if a country is abundant in factor j.
- Example: $F_k^i = K^i - s^i K^w$; $F_l^i = L^i - s^i L^w$.

2-3. Theorem due to Leamer (1980)

If capital is abundant relative to labor in country i, that is, $\frac{K^i}{K^w} > \frac{L^i}{L^w}$, then the HOV theorem implies that the capital-labor ratio embodied in production for country i exceeds the capital-labor ratio embodied in consumption:

$$\frac{K^i}{L^i} > \frac{K^i - F_k^i}{L^i - F_l^i}.$$

Proof

- Let capital and labor be the two production factors. By the HOV Theorem, we have $F_k^i = K^i - s^i K^w$, where we notice that the first element of V^i, V_k^i is the capital endowment K^i.

- This implies that $K^w = (K^i - F_k^i)/s^i$ and $\frac{K^i}{K^w} = \frac{s^i K^i}{K^i - F_k^i}$. Note that $K^i - F_k^i > 0$.

- Similarly, $L^w = (L^i - F_l^i)/s^i$ and $\frac{L^i}{L^w} = \frac{s^i L^i}{L^i - F_l^i}$.

- By the capital abundance assumption, we then have $\frac{K^i}{L^i} > \frac{K^i - F_k^i}{L^i - F_l^i}$. ■

2-4. Intuition behind the Leamer Theorem

- K^i is the capital endowment (calling it the capital embodied in production).

- F_k^i is the capital embodied in traded consumption (required to produce the traded amount of goods).

- $K^i - F_k^i$ is then the capital embodied in consumption (required to produce the consumption in country i).

- Similar for labor.

- Leamer Theorem says, the capital-labor ratio embodied in *production* must exceed that embodied in *consumption*, if the country is abundant in capital relative to labor.

3. Testing Heckscher-Ohlin-Vanek Model

- Testable equation: net tade flows satisfy $T^i = A^{-1}(V^i - s^i V^w)$.
- Testable predictions:
  - Sign test,
    $\text{sign}(F_k^i) = \text{sign}(V_k^i - s^i V_k^w), i = 1, \cdots, C; k = 1, \cdots, M;$
  - Rank test,
    $F_k^i > F_l^i \Leftrightarrow (V_k^i - s^i V_k^w) > (V_l^i - s^i V_l^w), i = 1, \cdots, C; k, l.$
    $= 1, \cdots, M$
- Many empirical tests. Overall, model performs poorly in fitting the actual data. For more, see Feenstra book pp.33-48.

- The main problems of Heckscher-Ohlin models lie in assumptions. In the real world
  - **Countries are not identical in technology.**
  - **Preference may be nonhomothetic.**
    - E.g., data show that the share of food in total consumption declines as income rises.

- **Technological differences across countries**
  - Total factor productivity differs significantly across countries, a key accounting source of income differences between countries.
  - Countries with scarce factor endowments may innovate to improve the productivity of these factors, leading to different factor intensities across countries.

- Empirical papers extended HOV Theorem to reflect technological differences, and find that doing so improves the model fit.
- **By doing that, these papers are no longer testing HOV Theorem. Rather they are also testing the Ricardian model.**